

Which Hong Kong Roads Are Most Sensitive to Rainstorms?

A Preprocessing and Integration Study of Detector, Rainfall and Warning Data

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Abstract—We join 332.9 million 30-second lane readings from about 790 Hong Kong road detectors with hourly district rainfall and rainstorm warning signals over three wet-season months (May 2024, July and August 2025, including all four 2025 Black Rainstorm episodes), through a three-layer pipeline whose every cleaning rule is recorded with evidence and counts. Speed falls steadily with the rain in the detector’s district (about 3.4 % per 10 mm/h), while the warning level mainly changes demand: under Black, flow is 62 % of normal whatever the local rain and speed is higher than under Red. A per-detector sensitivity ranking is stable across storms (split-half Spearman 0.84) but partly reflects how much rain each district received; after adjusting, Kowloon urban corridors are the most sensitive. Holding out whole storms, LightGBM reduces the error of a time-only baseline on wet slots by 28 %, almost all of it from rain features. An ablation of eleven preprocessing steps shows that the lane-to-detector speed aggregation, the time slot and the rain assignment change the answers, a robust outlier rule removes real congestion, and the validity flag, stuck sensors and gap filling hardly matter.

I. INTRODUCTION

Heavy rain is a recurring disruption in Hong Kong. The Observatory’s Black Rainstorm warning itself warns that persistent heavy rain “will cause serious road flooding and traffic congestion”, and 2025 was the first year under the current system with four Black Rainstorm episodes [1]. That rain slows traffic is well established; what a traffic manager or a commuter needs is *where* the effect is largest and whether it can be anticipated. Both questions require joining data that were never designed to be joined: 30-second lane readings from about 790 road detectors, an hourly rainfall *range* per district published as a weather bulletin, and warning signals issued for the whole territory.

This project treats that join, and the cleaning around it, as the object of study. We (i) build a reproducible three-layer pipeline (raw as written, cleaned per source, analysis table) over three wet-season months (May 2024, July and August 2025; 332.9 million lane readings), recording every cleaning rule with its evidence and counts; (ii) rank detectors by rainstorm sensitivity (RQ1); (iii) predict the 15-minute speed ratio during rain events against a time-only baseline, holding out whole storms (RQ2); and (iv) measure how much

each preprocessing choice changes the RQ1 ranking and the RQ2 accuracy (RQ3).

The main findings are: speed falls steadily with the rain *at the detector* (about 3.4 % per 10 mm/h), while the territory-wide warning level mostly changes *demand*: under Black the median flow is 62 % of normal whatever the local rain, and speed is higher than under Red at equal rain (a “Black paradox”). A warning-based sensitivity ranking is stable across storms (split-half Spearman 0.84) but half of its district pattern is *exposure* (how much rain fell there); after adjusting, the Kowloon urban corridors remain the most sensitive roads. Rain features cut the prediction error on wet slots by 28 % relative to the time-only baseline, while the warning level adds little. Of eleven preprocessing steps, the ones that change the answers are how lane readings become a detector speed (the ranking keeps only 11 of its top 20), the time slot (prediction skill 0.22 to 0.38) and how rain is assigned to a road; a robust outlier rule quietly removes a quarter of the rain congestion, and the validity flag, stuck sensors and gap filling hardly matter.

II. RELATED WORK

Rain and speed on a single road. A study of one Hong Kong urban road [2] reports speed reductions of about 4.2 %, 6.3 % and 7.3 % under light, moderate and heavy rain. *Rain and city-level congestion.* Work on Guangzhou and Shenzhen [3] models the nonlinear relation between daily rainfall, a city-wide congestion index and metro ridership. *The 2025 Black Rainstorms* are analysed meteorologically in [1]. Prior work thus uses either one road or a daily city-wide index; we work at detector level across the territory, at 15-minute resolution, conditioned on official warning levels, and we make the preprocessing itself the variable. We do not claim novelty for the finding that rain slows traffic.

III. DATA

All sources are free Hong Kong government open data (Table I). The study months are May 2024, July 2025 and August 2025: wet-season months with, together, 41 Amber (76 h), 18 Red (28 h) and 5 Black (21 h) rainstorm signals,

TABLE I
SOURCES USED (STUDY MONTHS ONLY)

ID	Content	Resolution / size
S1	TD detector readings [4]: speed, volume, occupancy, s.d., valid flag	30 s $\times$ lane; 784 detectors; 332.9M readings
S2	TD detector locations	district, road, lat/lon, direction
S3	HKO Current Weather Report [5]: past-hour rainfall range per district	hourly $\times$ 18 districts; 2,231 bulletins
S4	HKO rainstorm warning signals [6]	per signal (Amber, Red, Black)
S5	HKO tropical cyclone signals [6]	per signal (1–10)
S6	Public holidays	per date
S8	HKO daily rainfall	per day (checks only)

including all four 2025 Black episodes (29 Jul, 2 Aug, 4–5 Aug, 14 Aug), and Typhoon Wipha (signal 10, 20 Jul 2025).

S1 and S3 come from the DATA.GOV.HK Historical Archive, which stores a copy of the live file each time it fetched it. This shapes the data more than any sensor does. **Missing periods:** the archive did not fetch every minute; 27.0%, 46.6% and 41.0% of the 30-s periods are absent in the three months (at most 35 min in a row), so missing data are *absent rows*, not NA values. **Identical copies** of the same file are stored at several fetch times and must be deduplicated. **Midnight:** the 00:00:00 and 00:00:30 periods carry the previous day’s date. **Three truncated files** are byte-prefixes of complete copies fetched minutes apart. Inside S1, three quirks matter for the analysis: (a) with volume 0 the reported speed is the *speed limit* (70, 80, 100, 50 or 110 km/h in 99.9% of cases), not a measurement; (b) *valid* = N readings (2.9%) look normal, so only the flag identifies them; (c) speed 0 with volume > 0 occurs in 0.17% of readings with a mean of 11.3 vehicles per 30 s and occupancy 36%, physically impossible as a standstill; (d) speeds above 130 km/h (up to 300) occur in 0.56%. Volume is 0 in 20.2% of readings. In S3 the rainfall of a district is a range (e.g. “10 to 20 mm”) over several gauges, a district with no rain is simply not listed, and three hourly bulletins are missing.

IV. DESIGN AND IMPLEMENTATION

A. Pipeline overview

The pipeline has three layers, each written to Parquet and processed with DuckDB [7] (the S1 layer does not fit in the 16 GB of memory we used). **L1** parses every raw file *as written*: all values stay strings, nothing is trimmed, dropped or renamed, and structure checks (field sets, value forms, duplicates) run here. **L2** cleans each source separately by explicit rules; every rule has an ID, the evidence behind it, who decided it, and its row counts in a cleaning record. **L3** joins the sources into one analysis table, one row per detector and 15-minute slot; each of its steps has a default and alternatives (P1–P11) so that RQ3 can rebuild it with one choice changed (about one minute per build).

TABLE II
L2 RULES THAT CHANGE, ESTIMATE OR DROP DATA (THREE MONTHS)

Problem	Method	Rows
S1 00:00 periods dated the day before	re-date	329,465
S1 lanes listed twice in one block (AID02215)	mean	4 $\rightarrow$ 2
S1 direction absent for 4 new detectors	fill from later S1	268,072
S1 detector reporting a bad 3-lane layout (TDS90026, 2025)	drop	1,846 blocks
S1 Slow Lane absent (TDS90036)	ratio estimate	2,022 blocks
S1 lanes missing in a single block	drop	40 blocks
S1 occupancy -1 (volume 0)	set 0	7,687
S3 district not listed in bulletin	0 mm	33,340
S3 hourly bulletin missing	interpolate	3 hours
S3 same hour in two bulletins	keep one (equal)	2 hours
S5 summer-time flag	-1 h	451 signals

Two principles, agreed with the course teacher, guide L2. First, *fail loudly*: a case that no rule covers raises an error with the offending rows instead of being passed through or given a default, so a wrong assumption stops the pipeline where it is wrong. Second, *no NA in L2, with the method chosen by the kind of problem*: a few missing values with trustworthy neighbours are estimated (and flagged), a value given twice is averaged (and flagged), and rare bad data in a large source are dropped, after checking that the dropped rows do not cluster in rain periods.

B. L2 cleaning rules

Table II lists the rules that change data. Beyond these, L2 types every field, trims spaces (2.9M S1 readings had a trailing space in *direction*), and renames repeated lane names (Middle Lane twice in one detector, 98 k blocks) by position. Two rules came from checks failing: S1 occupancy = -1 occurs 7,687 times, always with volume 0 (set to 0); and three tropical-cyclone signals from the 1950s had a duration one hour longer than end minus start. All three start with an undocumented flag S and end with X; reading S as Hong Kong summer time (used until 1979) makes all 1,262 durations consistent, so S times are shifted to HKT.

The estimate for TDS90036 uses the detector’s own hour-of-day ratios from its complete blocks in the same month: the missing Slow Lane volume is the other lanes’ volume times the ratio of Slow to other-lane volume (1.2–5.9% by hour), speed the other lanes’ mean times the median speed ratio. Of the 1,846 dropped TDS90026 blocks, 0.9% (July) and 9.4% (August) fall in a rainstorm warning, against 4.3% and 8.9% of each month’s time, so the drop does not remove rain periods preferentially.

C. L3: the analysis table

Table III gives the L3 steps. Two defaults differ from our proposal, both for reasons found in the data. **P2**: the proposal filtered outliers with a per-detector robust z-score;

TABLE III
L3 STEPS: DEFAULT (AND RQ3 ALTERNATIVES)

P1	Drop lane readings with <code>valid = N</code> (keep; drop the whole detector period)
P2	Drop speed 0 or > 130 with volume > 0 (none; + robust $ z > 5$ per lane)
P3	Drop lane slots identical for ≥ 2 slots, 1,122 found (keep)
P4	Speed = $\sum v_i s_i / \sum v_i$; flow in veh/h; mean occupancy (mean of all readings; slowest lane)
P5	15-min slots (5 min; 60 min)
P6	Interpolate 1–2 missing slots, flagged, 25,787 (none)
P7	Own district, midpoint of range (upper bound; territory maximum)
P8	Rain of the hour the slot is in (the hour before)
P9	Level 0–3 and minutes since the episode began (any warning; none)
P10	Ratio to median of dry slots, same detector, season, day type, slot of day (per month; all months)
P11	Leave out TC signal ≥ 8 (keep)

but the slow-downs this project studies *are* outliers relative to a detector’s normal speed, so the default keeps only physical bounds and the z-score becomes an RQ3 alternative. **P10**: a baseline per month left 91 % of July 2025 Saturday slots without three dry reference slots (4 Saturdays, 2 under a cyclone signal); pooling July and August (both school summer holidays) and treating signals below 8 as dry fixes this. The speed of a slot is volume-weighted (P4), which gives the placeholder speeds of empty lanes weight 0.

The resulting table has 6,430,886 rows (784 detectors). The main quantity is the *speed ratio*: the slot’s speed divided by the detector’s dry-weather median for the same season, day type (weekday, Saturday, Sunday or holiday) and time of day. The flow ratio is defined in the same way. A slot is *dry* if its district had no rain in that hour and the hour before, no rainstorm warning and a cyclone signal below 8.

D. Methods

RQ1. Per detector, s_{warn} is the median speed ratio under Red or Black minus 1 (Red and Black pooled; Black alone is four episodes) and s_{rain} the same with district rain ≥ 10 mm/h. Because districts received different amounts of rain, we also compute an exposure-adjusted s_{warn} (removing the linear dependence on the median district rain while Red/Black was in force) and a per-detector rain slope from a mixed model, ratio $\sim \text{rain} + (1 + \text{rain} | \text{detector})$, on 300k wet slots. Stability is tested by splitting the rain events into two halves. **RQ2.** On the 42 event days (a warning or ≥ 10 mm/h somewhere; consecutive days merged into 22 events) we predict the speed ratio and a congestion flag (ratio < 0.7). The baseline predicts ratio 1 (the dry profile for that detector and time); we compare ridge / logistic regression and LightGBM [8] with four feature sets. Folds are grouped by event, so a storm is never in both training and test data. **RQ3.** Each alternative in Table III is applied alone; we report the Spearman correlation of s_{warn} with the default ranking, the overlap of the 20 most sensitive detectors, the headline ratios, and the LightGBM

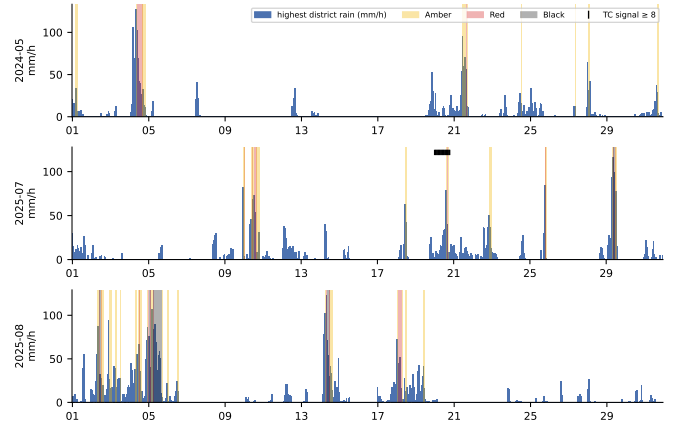


Fig. 1. Highest district rainfall per hour and rainstorm warnings (shaded Amber, Red, Black) in the three months; ticks mark cyclone signal ≥ 8 .

TABLE IV
MEDIAN RATIOS TO THE DRY BASELINE BY WARNING LEVEL (15-MIN DETECTOR SLOTS)

Warning	Slots	Speed	Speed P10	Flow
None	5,988,953	0.999	0.940	1.000
Amber	217,036	0.940	0.817	0.930
Red	75,776	0.918	0.770	0.902
Black	66,391	0.937	0.785	0.620

skill 1 – $\text{MAE}/\text{MAE}_{\text{baseline}}$ (the target itself changes with the variant, so skill is the comparable measure).

V. EVALUATION

A. Exploration: rain, warnings, speed and flow

Fig. 1 shows the rain and warnings of the three months. Table IV and Fig. 2 summarise the speed and flow ratios (cyclone signal ≥ 8 left out). Speed falls monotonically with the rain in the detector’s district: without a warning from 0.976 at 0–5 mm/h to 0.888 above 40 mm/h, and under a warning from 0.951 to 0.825. By warning level, however, the ordering breaks: the median speed ratio is 0.940 under Amber, 0.918 under Red but 0.937 under Black. The flow ratio explains why: it is 0.93 and 0.90 under Amber and Red but 0.62 under Black, and it stays near 0.62 in every rain bin, even where the district had no rain at all (where the Black speed ratio is 1.007, faster than normal). At equal local rain, speed under Black is never below that under Amber or Red up to 40 mm/h, and above Red in every bin (Fig. 2, left); only above 40 mm/h is Black the slowest.

The 5 August 2025 Black Rainstorm (Fig. 3) shows the mechanism in time. The Black signal was issued at 05:50; at 06:00 the median speed ratio reached its minimum of 0.77 while the morning traffic was already on the road. Flow then collapsed to 0.38 of normal by about 09:00 and speed recovered to 1.0 by 16:00, while Black was still in force. After the signal was cancelled at 17:05, flow stayed near half of normal through the evening. We read this as a demand response to the Black signal (under the Labour Department’s

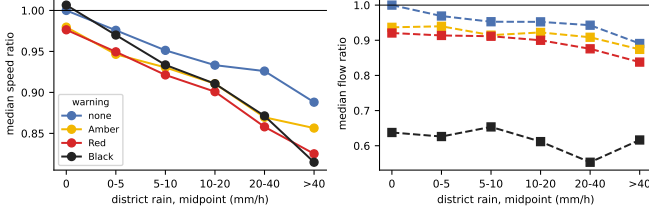


Fig. 2. Median speed ratio (left) and flow ratio (right) by district rain and warning level. At equal local rain the warning level changes flow far more than speed.

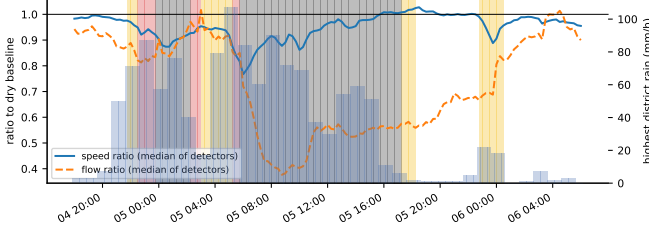


Fig. 3. 5 August 2025: median speed and flow ratio over detectors, highest district rain (bars) and warning level (shading), from 18:00 on 4 August to 06:00 on 6 August.

code of practice, employees are generally not required to report for work [9]); see Section VI for the hypotheses and checks.

B. RQ1: sensitivity ranking

781 detectors have at least 20 Red/Black slots. Their s_{warn} has median -0.075 (interquartile range -0.114 to -0.044); the most sensitive detector falls by 21%. The ranking is a stable property of the road: s_{warn} and s_{rain} correlate at $\rho = 0.93$, and computed on two disjoint halves of the rain events the rankings correlate at 0.84 (s_{warn}) and 0.90 (s_{rain}). But s_{warn} also correlates at -0.51 with the rain the detector’s district actually received while Red or Black was in force (median 5 mm/h in North District, 14.5 mm/h on the Islands). Half of the “district pattern” of a warning-based ranking is therefore exposure, not sensitivity. After adjusting (Fig. 4), the Kowloon urban districts (Yau Tsim Mong, Kowloon City, Sham Shui Po, Kwun Tong) remain the most sensitive; North District moves up and the Islands move down (their expressways had the most rain but fall least per mm). 13 of the 20 most sensitive detectors are the same before and after adjustment; they are urban corridors and their approaches: Tseung Kwan O Road, Kwun Tong Bypass, Lung Cheung Road, West Kowloon Corridor, Chatham Road North and Lion Rock Tunnel Road. The mixed model gives a mean slope of -0.034 speed ratio per 10 mm/h (s.e. 0.001) with a between-detector standard deviation of 0.035: the slope itself varies as much as its mean. Baseline occupancy correlates positively with s_{warn} ($\rho = +0.25$): busier roads lose less of their normal speed.

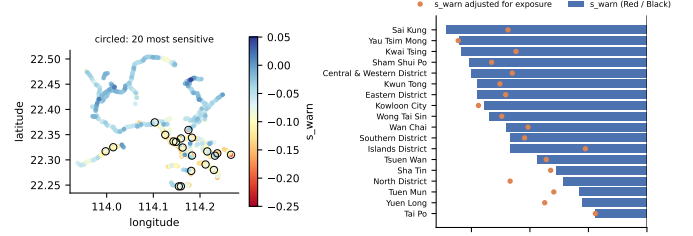


Fig. 4. Left: s_{warn} per detector (circled: 20 most sensitive). Right: median per district, raw and adjusted for rain exposure.

TABLE V
RQ2: MEAN OVER 5 EVENT-HELD-OUT FOLDS (MAE OF SPEED RATIO; F1 OF RATIO < 0.7)

Model	Features	MAE wet	MAE R/B	F1 R/B
Baseline	ratio = 1	0.0792	0.1113	0.000
Ridge / logistic	+rain+warning	0.0638	0.0818	0.091
LightGBM	time+road	0.0731	0.1033	0.164
LightGBM	+rain	0.0574	0.0725	0.157
LightGBM	+warning	0.0665	0.0846	0.137
LightGBM	+rain+warning	0.0568	0.0717	0.167

C. RQ2: prediction during rain events

The 42 event days hold 2.83 M slots; 1.1 % are congested (ratio < 0.7), 3.9 % under Red/Black. Table V gives the five-fold, event-held-out results. LightGBM with rain and warning features reduces the MAE on wet slots from 0.079 (baseline) to 0.057, a 28 % reduction, and on Red/Black slots from 0.111 to 0.072 (−36 %). The feature sets separate the two kinds of information: rain features alone reach 0.057, warning features alone only 0.067, and adding warnings to rain changes the MAE by 0.0006. In the gain importance the district rain of the hour (35 %) and of the hour before (24 %) dominate; the warning level has 2 %. Congestion as a class is hard: F1 stays at 0.14–0.17 for all models, far below what the regression gain would suggest.

D. RQ3: how much does preprocessing matter?

Each of the 18 alternatives in Table III was run with every other step at its default: the L3 table is rebuilt (75–160 s each), RQ1 is recomputed, and LightGBM with all features is refitted on folds fixed by event and 1 M training slots per fold. Because the target itself changes with the variant, we compare *skill* ($1 - \text{MAE}/\text{MAE of the ratio} = 1$ baseline on the same slots) rather than raw MAE. Table VI and Fig. 5 give the results. Variants whose RQ2 data are unchanged (P6: RQ2 already leaves interpolated slots out) give skill 0.277–0.280, so differences below about 0.005 are noise from subsampling.

How speed is measured matters most. The lane-to-detector aggregation (P4) is the only step that reorders the ranking substantially: with either alternative only 11 of the 20 most sensitive detectors remain. Averaging all readings lets the empty-lane readings, whose “speed” is the posted limit, dilute every ratio towards 1 (Black 0.961 instead of 0.937) and

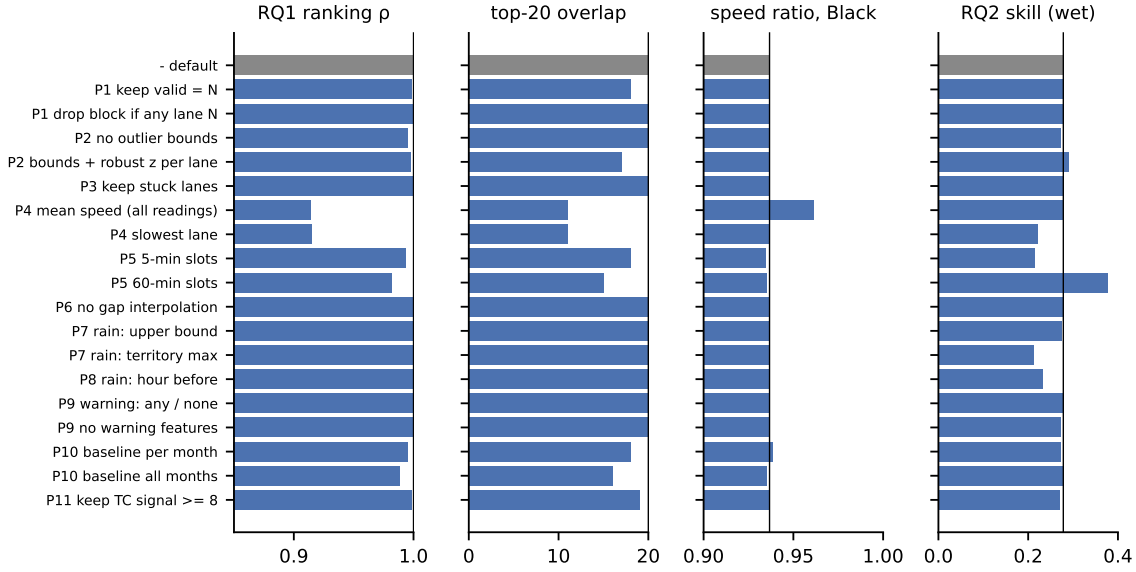


Fig. 5. RQ3: each alternative against the default (grey bar, black line).

TABLE VI

RQ3: ONE PREPROCESSING ALTERNATIVE AT A TIME. ρ : SPEARMAN OF s_{warn} WITH THE DEFAULT OVER 781 DETECTORS; TOP-20: OVERLAP OF THE 20 MOST SENSITIVE; BLACK: MEDIAN SPEED RATIO; CONG.: SHARE OF RED/BLACK SLOTS WITH RATIO < 0.7 ; SKILL AND F1 AS IN RQ2

Variant	ρ	top-20	Black	cong.	skill	F1
Default	1.000	20	0.937	3.9 %	0.278	0.168
P1 keep valid=N	0.999	18	0.937	4.0 %	0.280	0.159
P1 drop period if any N	1.000	20	0.937	3.9 %	0.280	0.161
P2 no bounds	0.995	20	0.936	4.2 %	0.273	0.165
P2 bounds + robust z	0.998	17	0.937	3.0 %	0.290	0.138
P3 keep stuck lanes	1.000	20	0.937	3.9 %	0.279	0.165
P4 mean of readings	0.914	11	0.961	1.7 %	0.277	0.124
P4 slowest lane	0.915	11	0.936	6.2 %	0.221	0.210
P5 5-min slots	0.994	18	0.934	4.9 %	0.215	0.172
P5 60-min slots	0.982	15	0.935	2.6 %	0.378	0.148
P6 no interpolation	1.000	20	0.937	3.9 %	0.279	0.161
P7 rain upper bound	–	20	0.937	3.9 %	0.274	0.162
P7 territory maximum	–	20	0.937	3.9 %	0.212	0.137
P8 rain of hour before	–	20	0.937	3.9 %	0.233	0.144
P9 warning any/none	–	20	0.937	3.9 %	0.277	0.155
P9 no warning features	–	20	0.937	3.9 %	0.272	0.144
P10 baseline per month	0.995	18	0.939	3.9 %	0.273	0.159
P10 baseline all months	0.989	16	0.935	3.9 %	0.280	0.164
P11 keep TC signal ≥ 8	0.998	19	0.937	3.9 %	0.272	0.158

–: RQ1 does not use the changed feature (it uses the midpoint rain of the hour and the warning level), so the ranking is unchanged by construction.

halves the congested share; taking the slowest lane does the opposite (6.2 % congested, higher F1, lower skill). *The time resolution changes what can be predicted*: skill is 0.215 at 5 min, 0.278 at 15 min and 0.378 at 60 min, because the rain input is hourly and a longer slot averages out the variation that rain cannot explain; the ranking still keeps 15 of 20. *The rain assignment matters for prediction only*: the territory maximum instead of the own district’s rain lowers skill by a quarter (0.278 to 0.212), and the rain of the hour before instead of

the current hour lowers it to 0.233; the upper bound of the range is as good as the midpoint. Warning features, by contrast, change skill by 0.006 when removed, though they help the congestion classifier (F1 0.168 to 0.144). *An outlier rule can remove the phenomenon*: the robust z rule (P2) raises skill to 0.290, but the 1.73 M readings it adds to the exclusions are mostly slow (72 % below the lane median; median 20 km/h on lanes with a median of 72 km/h) and over-represented under Red and Black (1.7 times their share of the kept readings), so it removes a quarter of the congestion under Red/Black (3.9 to 3.0 %) and F1 falls. *The steps most discussed during cleaning matter least*: the validity flag (P1), stuck sensors (P3), gap interpolation (P6) and the cyclone filter (P11) all keep $\rho \geq 0.998$ and change skill by less than the noise. The baseline (P10) is in between: a single baseline for all three months keeps 16 of the top 20.

VI. DISCUSSION

Several results differ from what we expected. We state them as hypotheses with the check that would confirm or reject each; none is verified yet.

H1: the Black paradox is a demand effect. Under Black the speed ratio is higher than under Red because fewer vehicles travel: flow is 0.62 of normal in every rain bin, including districts without rain, and keeps falling with time since the signal (0.95 in the first 30 min, 0.52 after 8 h). *Against*: there are only four Black episodes, three on weekdays, and 5 August dominates. *Check*: per-episode curves; weekday vs weekend episodes; a regression of speed ratio on rain, flow ratio and level.

H2: the warning level is a territory-wide label; local rain drives speed. At equal district rain the levels differ by at most 0.03 in speed ratio below 40 mm/h, and in RQ2 the warning features add almost nothing to rain features. RQ3

agrees: replacing the district's rain by the territory maximum, which is what a territory-wide signal summarises, lowers skill by a quarter. *Check*: rain of neighbouring districts and of the route; the radar nowcast at the detector.

H3: the speed shock is at warning onset. On 5 August the minimum speed ratio came 10 minutes after the Black signal, and speed recovered while Black was still in force. *Against*: pooled over all episodes the speed ratio is flat in time since issue (Amber 0.94 throughout). *Check*: episode curves aligned at issue time, weekday mornings only.

H4: a warning adds a slow-down beyond the district rain (at 20–40 mm/h: 0.926 without, 0.870 with a warning): either rain elsewhere on the route, or the hourly district range misses short intense bursts. *Check*: add neighbouring districts' rain; S7 radar nowcast at 6-min resolution.

H5: sensitive roads are near capacity, yet busier roads lose less in ratio ($\rho = +0.25$ with baseline occupancy). A ratio has less room to fall where the dry baseline is already slow, and demand relief helps busy roads most. *Check*: rank on the speed difference in km/h instead of the ratio; road class from the road network data.

H6: congestion is local. Speed ratio < 0.7 is predicted with F1 0.14–0.17 although the MAE improves by 28 %; incidents, flooded lanes or closures would explain this. *Check*: join the Transport Department's special traffic news to the congested slots.

H7: stuck presence readings mark flooding or closures. All 1,122 lane slots removed by P3 read occupancy 100 with speed and volume 0; the day with most is 20 July 2025, under cyclone signal 10 (10 detectors). *Check*: traffic news for those roads.

H8: robust outlier rules delete rain congestion. The z rule's extra exclusions are slow readings concentrated in warning periods (Section V-D). A per-lane distribution over three months treats a real queue as an outlier. *Check*: compare the excluded readings with the neighbouring lanes and detectors in the same 30 s; a conditional rule (outlier relative to the same slot of day, or only when the neighbours disagree).

H9: the P4 alternatives reorder the ranking through the empty-lane placeholder and lane mix. Mean-of-readings dilutes by the share of empty lanes, which is highest on quiet roads and at night; slowest-lane follows the slow or exit lane. *Check*: ranking with a volume-weighted mean restricted to the same lanes over time; per-lane sensitivity.

H10: the gain at 60 min is resolution matching, not a better model. Rain is an hourly district range, so at 15 and 5 min the target has variation the input cannot see. *Check*: 15-min targets with the 6-min radar nowcast (S7); skill at 60 min without rain features.

H11: speed 0 with vehicles counted is a sensor fault, not a standstill. These readings (0.17 %, left out by P2) carry a mean of 11.3 vehicles per 30 s and 36 % occupancy, which cannot pass at 0 km/h; in May 2024 half of them come from one detector (TDS90023). *Check*: the neighbouring lanes in the same 30 s and the detector's speed in the minutes before and after.

H12: a warning-based ranking mixes sensitivity with exposure. s_{warn} correlates at -0.51 with the rain the district received under Red/Black, although the ranking is stable across storms (split-half 0.84). *Check*: compare the rankings by s_{warn} , the exposure-adjusted s_{warn} and the mixed-model slope on more events.

Limitations. Three months, four Black episodes; rainfall is an hourly range per district, not a gauge at the road; the archive misses 27–47 % of the 30-s periods; the baseline assumes the dry days of a season are comparable; we have no incident data in the model.

VII. CONCLUSION

Joining detector, rainfall and warning data at 15-minute detector level shows that where it rains matters more than which signal is hoisted: speed follows the district's rain, the warning level mostly changes how many people travel, and a sensitivity ranking built on warnings must be adjusted for exposure before it says anything about the road. With that adjustment the Kowloon urban corridors stand out. Rain features make storm-time speed predictable in aggregate, but congestion episodes remain largely unexplained. For the preprocessing, the lesson of the ablation is that the consequential choices are those that define the quantity being measured (speed aggregation, slot length, rain assignment, what counts as an outlier), not the ones that decide which few readings to keep. The next steps are the checks listed with H1–H12: per-episode analysis of the Black paradox, traffic news for congestion and stuck sensors, radar rainfall at the detector, and more months.

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[Contribution of each group member: to be written by the group.]

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